

StoreMore optimization tool description

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Co-funded by
the European Union

StoreMore

Energy system modelling software

Name of scenario:

Include:

☐ RPS constraint
Apply minimum renewable share requirement for generation

☐ CO₂ constraint
Apply maximum CO₂ emission limits during optimization

☐ CEEP constraint
Limit Critical Excess Electricity Production to allowed level

☐ Optimize generation capacity
Allow capacity expansion of generation units

☐ Restrict investments
Restrict investments into generation capacities, storage solutions or P2H

Enable storage technologies:

☐ Li-ion storage
Model lithium-ion battery storage systems

☐ Use ratio for Li-ion storage
Apply predetermined ratio for Li-ion

☐ Gravity storage
Model gravitational energy storage systems

☐ Use ratio for Gravity storage
Apply predetermined ratio for Gravity

Data:

Modify Generator Parameters

Unit Name	Installed Capacity [MW]	Max Investment [MW]	Efficiency [share]	Capital Investment Cost [M EUR/MW]	CO ₂ intensity [tCO ₂ /MWh]
PP gas	0.0	0.0	0.31	1.3	0.398
solar	100.0	0.0	—	0.56	—
wind	20.0	0.0	—	1.1	—

Storage Tech Data

Unit Name	Max Investment [MWh]	Installed Capacity [MWh]	Efficiency [-]	Storage Ratio [-]	Variable Cost [EUR/MWh]	Input/output capacity [MW]	Maximum investment into input/output system [MW]	Hourly storage loss as a share of SOC [-]
Lion storage	0.0	2.0	0.9	0.5	20.0	0.0	0.0	2e-05
Gravity storage	0.0	2.0	0.84	0.17	15.0	0.0	0.0	1e-06
CAES	0.0	0.0	0.51	0.07	30.0	0.0	0.0	1e-06
LAES	0.0	0.0	0.6	0.5	80.0	0.0	0.0	0.0001
VRFB	0.0	0.0	0.75	0.1	60.0	0.0	0.0	1e-05
SMES	0.0	0.0	0.925	360.0	55.0	0.0	0.0	0.001
Flywheel storage	0.0	0.0	0.9	360.0	55.0	0.0	0.0	0.03
Ultracapacitor	0.0	0.0	0.92	360.0	55.0	0.0	0.0	0.003

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1. Overview

StoreMore is a web-based tool for modeling, simulating, and optimizing future energy systems, focusing on electricity demand, generation, sector coupling, and storage technologies. It supports technology-specific constraints, investment planning, and energy balance evaluations.

The tool is intended for local policy makers and operators of energy storage and or renewable power generation systems. The outputs from the tool enable these stakeholders to bring decisions on investments into storage systems and into generation from renewable sources.

The basic structure is displayed on Figure 1. Non-dispatchable renewable sources like wind or solar PV can generate electricity and feed it into the grid. There is also the possibility of having the existing thermal power plant as well as the interconnection to the remainder of the power system via import/export capacity. Depending on the prices in the market, the model can choose to import or export electricity if the interconnection is available. In order to balance the supply and demand, the model can utilize energy storage systems as well. Unutilized but available energy will be called excess production if it cannot be used to satisfy the demand, fill the storages or export, but in reality, this would be curtailed production.

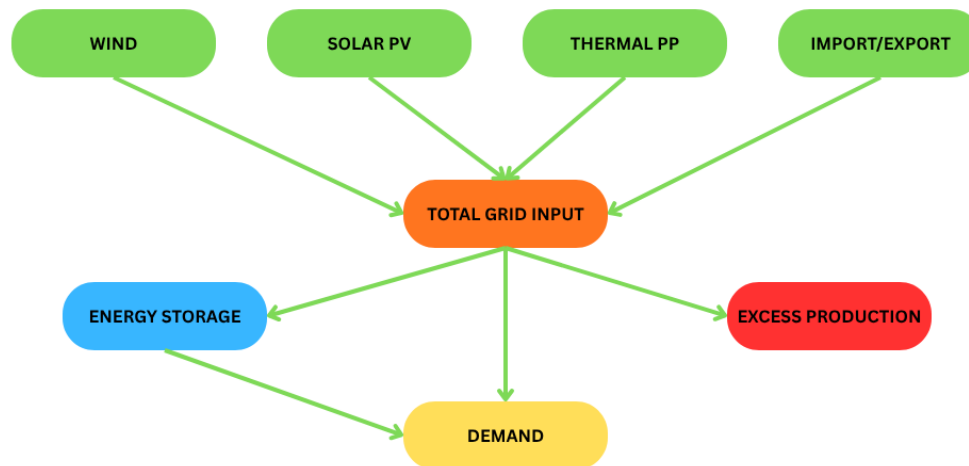


Figure 1. Energy structure of the model

Current configuration:

The optimization gives the results and runs for 1 entire year or 8760 hours, but the effects during the entire technology lifespan are implemented in order to account for the disparity of CAPEX and OPEX if only a single year is modelled.

The model is run in two stages. The first stage includes the optimization of the entire planning horizon which is in this case the entire year. This stage gives the results on the necessary investments to optimize the system if the optimization function is enabled. Then, the results on the investments are saved and the second stage begins. This stage consists of a range of 365 48 hour long partially overlapping optimization tasks. The planning horizon of each individual optimization step is divided into 2 parts. The first 24 hours use historical data from the tables for the distributions of electricity demand, renewable energy resource availability, electricity prices and heat demand. The second 24 hours use the predicted data that is based on the same distributions but has the uncertainties embedded. This replicates the uncertainties in the forecasts, and the system adapts to the slightly modified version of reality. Then, in the next step, it is faced with the real data and has to adapt. In order to generate the predicted data, neural networks model was used.

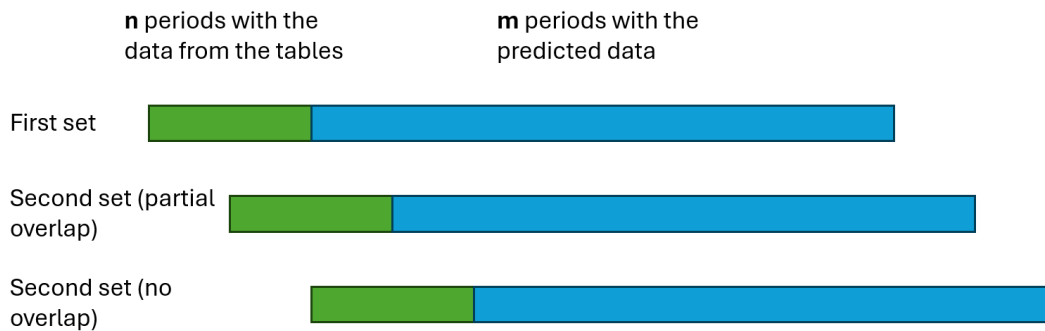


Figure 2. Overlaps in the optimization stages

The model is available online via the user interface that is designed to be intuitive and user friendly. The interface is easily accessible from any location or device that can open basic html web pages. This includes both computers as well as the use on mobile devices. The only requirement is that the user does not refresh the website since in that case communication with the process at hand will be lost. The optimization tool on mobile phone browser is displayed on Figure 3.

afdedb254.ngrok-free.app

StoreMore
Energy system modelling software

Name of scenario:

Include:

RPS constraint
Apply minimum renewable share requirement for generation. ☐

CO₂ constraint
Apply maximum CO₂ emission limits during optimization. ☐

CEEP constraint
Limit Critical Excess Electricity Production to allowed level. ☐

Optimize generation capacity
Allow capacity expansion of generation units. ☐

Restrict investments
Restrict investments into generation capacities, storage solutions or PSH. ☐

Enable storage technologies:

Li-ion storage
Model lithium-ion battery storage systems. ☐

Use ratio for Li-ion storage
Apply predetermined ratio for Li-ion. ☐

Gravity storage
Model gravitational energy storage systems. ☐

Use ratio for Gravity storage
Apply predetermined ratio for Gravity. ☐

CAES
Model Compressed Air Energy Storage. ☐

Use ratio for CAES
Apply predetermined ratio for CAES. ☐

LAES
Model Liquid Air Energy Storage. ☐

Use ratio for LAES
Apply predetermined ratio for LAES. ☐

VRFB
Model Vanadium Redox Flow Batteries. ☐

Use ratio for VRFB
Apply predetermined ratio for VRFB. ☐

SMES
Model Superconducting Magnetic Energy Storage. ☐

Use ratio for SMES
Apply predetermined ratio for SMES. ☐

Flywheel storage
Model Flywheel storage. ☐

Use ratio for Flywheel storage
Apply predetermined ratio for Flywheel storage. ☐

Ultracapacitor
Model ultracapacitor storage. ☐

Use ratio for Ultracapacitor storage
Apply predetermined ratio for Ultracapacitor storage. ☐

Model Parameters:

Total electricity demand [MWh]
Total annual electricity demand used for scaling.

Service life [years]
Lifetime of investment technologies.

No-load electricity cost [€/MW]
Fixed electricity cost per MW capacity (e.g. maintenance).

Import/export capacity [MW]
Transmission capacity available for imports and exports.

Data:

Modify Generator Parameters

Unit Name	Installed Capacity [MW]	Max Investment [MW]	Efficiency [share]	Capital Investment Cost [M EUR/MW]	CO ₂ Inter [tCO ₂ /MWh]
PP	0.0	0.0	0.31	1.3	0.398
gas	100.0	0.0	—	0.56	—
solar	20.0	0.0	—	1.1	—
wind					

Storage Tech Data

Unit Name	Max Investment [MWh]	Installed Capacity [MWh]	Efficiency [-]	Storage Ratio [-]
Lion storage	0.0	0.1	0.9	0.5
Gravity storage	0.0	2.0	0.84	0.17
CAES	0.0	0.0	0.51	0.07
LAES	0.0	0.0	0.6	0.5
VRFB	0.0	0.0	0.75	0.1
SMES	0.0	0.0	0.925	360.0
Flywheel storage	0.0	0.0	0.9	360.0
Ultracapacitor	0.0	0.0	0.92	360.0

Annual Investment Cost

Technology	Investment Cost	Unit	Description
Lion storage	1.042	M EUR/MWh	Lithium-ion battery storage
Gravity storage	3.82	M EUR/MWh	Gravity-based energy storage (e.g., lifting masses)
CAES	9.0	M EUR/MWh	Compressed Air Energy Storage
LAES	1.17	M EUR/MWh	Liquid Air Energy Storage
VRFB	20.0	M EUR/MWh	Vanadium Redox Flow Battery
SMES	1.15	M EUR/MWh	Superconducting Magnetic Energy Storage
Flywheel storage	1.15	M EUR/MWh	Flywheel energy storage
Ultracapacitor	1.15	M EUR/MWh	Ultracapacitor energy storage
Lion storage in out	0.1042000000	M EUR/MW	Power conversion and grid interface for lithium-ion storage (inverters and PCS), per MW.
Gravity storage in out	0.382	M EUR/MW	Hoisting and motor-generator equipment for gravity storage, including drivetrain and power electronics, per MW.
CAES in out	0.9	M EUR/MW	Compressors, expanders, motor-generators, and power electronics, per MW.

Figure 3. Optimization tool in mobile phone browser

2. Available model settings and functionalities

- RPS constraint – Minimum RES (Renewable Energy Source) share requirement in the power generation. When activated, this constraint will force the model to satisfy the minimum defined share of renewable energy in power generation. This will result in investments in wind and solar systems.

- CO₂ constraint – Limit on total annual CO₂ emissions in the system. Similarly, as in RPS constraint, this will result in investments into solar and wind power with the possible exception if the interconnection capacity is available.

- CEEP constraint – This option limits the “wasteful” power generation meaning that majority of the generated electricity has to be utilized or in other words to minimize curtailment. Activation of this option will force the model to utilize energy storage to balance out generation and demand profiles. Restricts Critical Excess Electricity Production.

- Optimization of generation capacity as well as the storage capacity – When this option is active, the model will provide the results on the required capacity expansion in the generation and storage systems.

- Restrict investments – It is possible to exogenously prevent the model in investments that may be above the ones that are allowed by the geographical and zoning constraints in the modelled region.

The model outputs the results on the necessary investments into both power generation and energy storage systems. Also, total investment cost is displayed. Besides the investments, the results display the optimized dispatch which includes the strategies on charging and discharging the storage systems.

3. Input data and settings

In this section, very basic information on model setup is given while more extensive information is available in the user manual.

The model is compatible with the community to city scale simple power grids. In the inputs of the model, the users can define the existing capacities and basic technology data as efficiency of the generators as displayed on Figure 4.

Modify Generator Parameters

Unit Name	Installed Capacity [MW]	Max Investment [MW]	Efficiency [share]	Capital Investment Cost [M EUR/MW]	CO ₂ Intensity [tCO ₂ /MWh]
PP gas	0.0	0.0	0.31	1.3	0.398
solar	100.0	0.0	—	0.56	—
wind	20.0	0.0	—	1.1	—

Figure 4. Data inputs for the generators

The distribution curves both for power demand, day ahead electricity prices and availability of renewable sources is determined by selecting the modelled country from the drop-down menu as displayed on Figure 5. It is also possible to upload custom electricity price distributions instead of using the ones from the database.

Country code

Two-letter ISO country code (e.g. HR, DE, FR)

SE
HR ▼

Figure 5. Country selection

Users can also define the starting capacities in the storage systems as it is displayed on Figure 6. Besides starting capacities, the technical data such as charging efficiency, self-discharge rate, ratio of input/output to the storage capacity and investment data can be specified.

Storage Tech Data

Unit Name	Max Investment [MWh]	Installed Capacity [MWh]	Efficiency [-]	Storage Ratio [-]	Variable Cost [EUR/MWh]	Input/output capacity [MW]	Maximum investment into input/output system [MW]	Hourly storage loss as a share of SOC [-]
Liion storage	0.0	2.0	0.9	0.5	20.0	0.0	0.0	2e-05
Gravity storage	0.0	2.0	0.84	0.17	15.0	0.0	0.0	1e-06
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VRFB	0.0	0.0	0.75	0.1	60.0	0.0	0.0	1e-05
SMES	0.0	0.0	0.925	360.0	55.0	0.0	0.0	0.001
Flywheel storage	0.0	0.0	0.9	360.0	55.0	0.0	0.0	0.03
Ultracapacitor	0.0	0.0	0.92	360.0	55.0	0.0	0.0	0.003

Figure 6. Data inputs for storage technologies

4. Running the optimization

The model is available at this link. Start by opening the link in your browser.

<https://56a47ef58d5d.ngrok-free.app>

Click Start simulation to run the model with current inputs.

Click Quit to exit without running.

The interface will display in which phase of the process the model is. In the second phase, when solving consecutive steps, the progress bar is visible.

After the model is complete, the link to download the results will appear. Only .png and .svg files are downloaded.

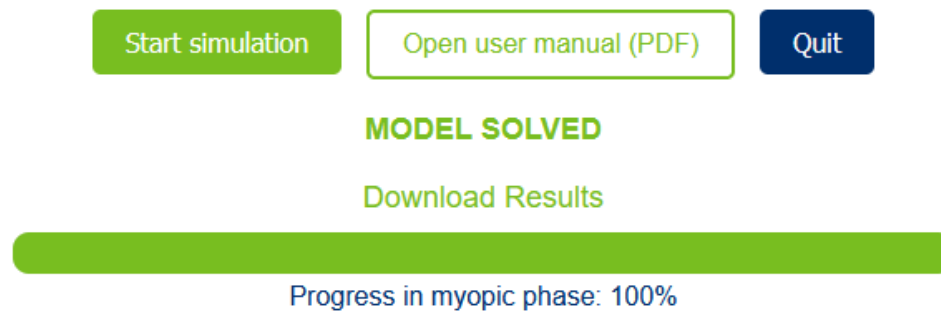


Figure 7. Model status

5. Examples of the results and the process of optimization

When clicked on “start simulation”, the window will change to display “solving first phase – perfect foresight whole year” to indicate that the process is in the first phase. This appears only if optimization of the capacities is enabled. If it is not, the process will skip this step and go directly to the myopic phase. In this phase, the whole year was modelled and optimized at once. After running this phase, the results on the capacity investments were extracted and they will be used in the myopic phase. The capacity investments are constrained in the case that the option to use investment constraints is active. In that case, the undelivered electricity is possible, but it will be penalized.

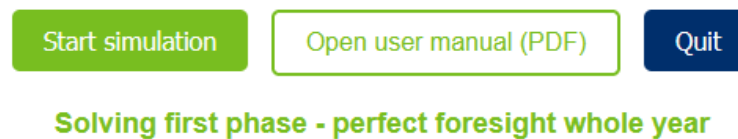


Figure 8. Perfect foresight phase

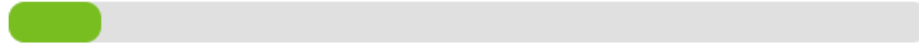
After finishing the perfect foresight phase, the model will transition to the myopic phase. Here, the previously captured investments will be exogenously implemented. In this phase, 48-hour segments are run sequentially with 24-hour overlap. The first 24-hour datasets are using the real data while the next 24-hour segment is using the predicted data in order to replicate uncertainty in day ahead and in the weather forecasts.

Start simulation

Open user manual (PDF)

Quit

MODEL RUNNING



Progress in myopic phase: 10%

Figure 9. Myopic phase

After the myopic phase is finished, the results are available for download. They include only the graphical results.

MODEL SOLVED

Download Results



Progress in myopic phase: 100%

Figure 10: End of myopic phase

Example of the power generation results are displayed in Figure 11. On this example that included the closed system with the capacities in wind, solar and gas thermal power. It also included the possibility of expanding the capacity of wind and solar as well as investing into stationary battery storage systems. On the positive Y axis, the power generation from wind and solar are visible as well as the output from the stationary storage system. On the negative axis, the charging of the storage is visible. Also, the matching between the availability of power generation and inputs to the storage are visible.

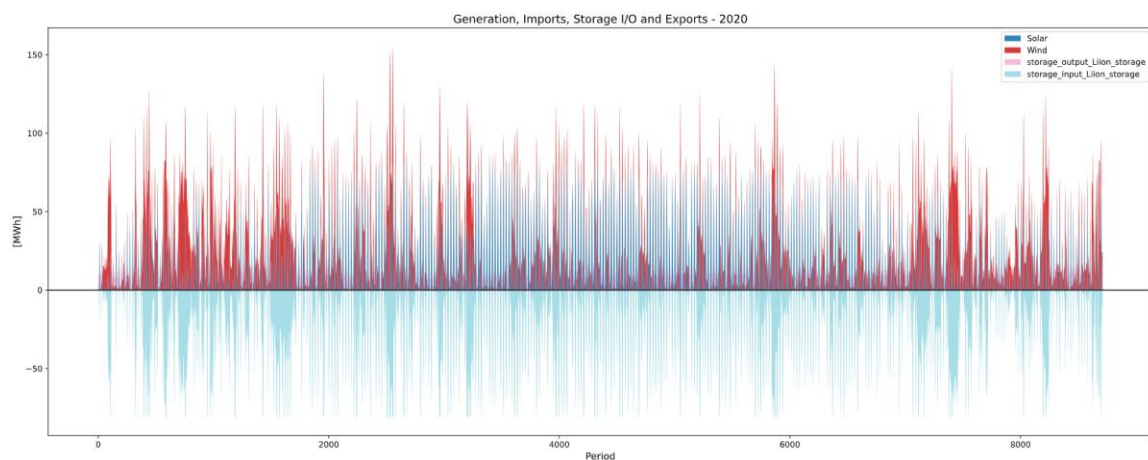


Figure 11. Power generation

Capacity expansion in power generation is displayed in Figure 12.

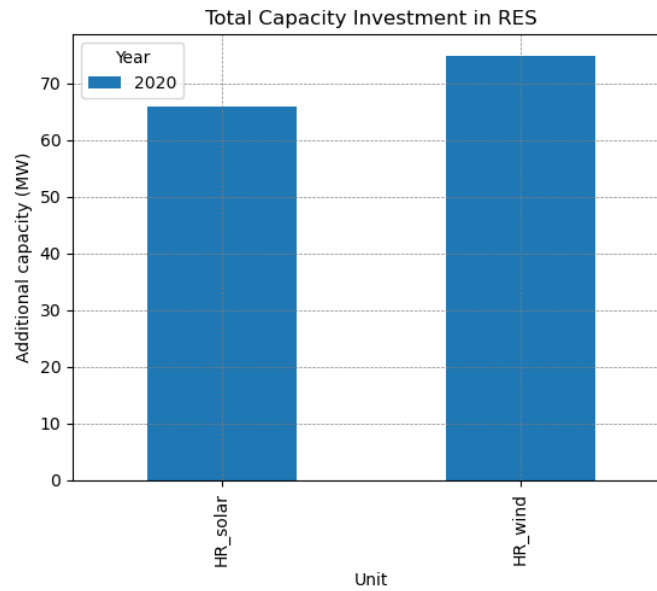


Figure 12. Investments into generation capacities

The investments into stationary storage systems are displayed on Figure 13.



Figure 13. Investments into storage capacities

Input and output to the storage are visible in Figure 14. The charging is performed mostly at higher capacities when the power generation is available for shorter periods of time while the discharging is performed more gradually with smaller output capacities.

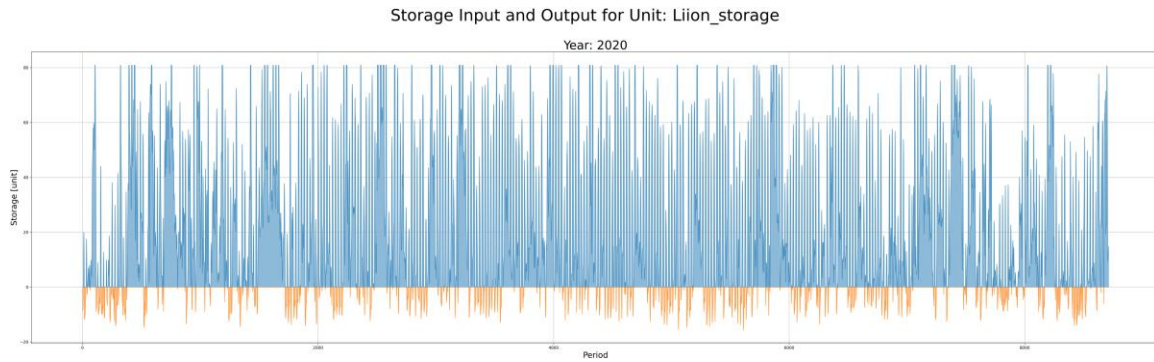


Figure 14. Input/output to storage

The state of charge of the system is displayed on Figure 15.

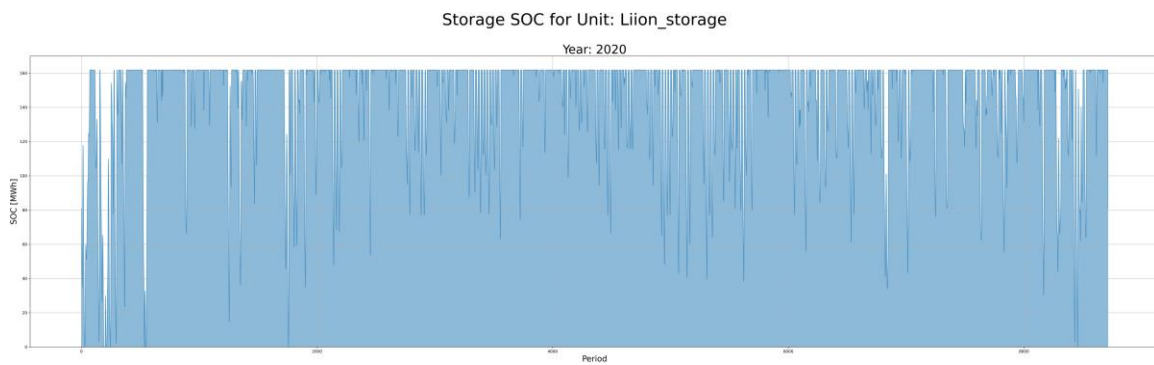


Figure 15. Storage SOC